SINGAPORE DIMENSIONS • KINDERGARTEN (KA) • CHAPTER 1

Numbers to 5: Counting & Cardinality

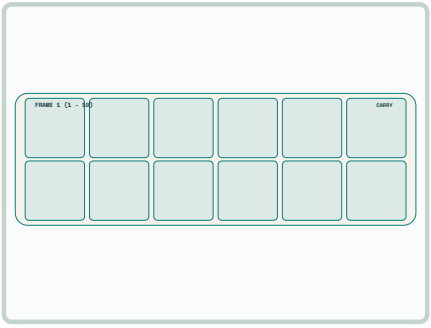
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.K.CC.B.4

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Ten Frame Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place two-color counters in the top row of the 2x5 frame to represent 1 to 5.

Sketch Model on Paper:
Color in circles in a 5-frame grid on the worksheet.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

1.	4	X: 4	Key 1st operand
2.	ENTER	Y: 4 | X: 4	Stack lift copies X
3.	1	Y: 4 | X: 1	Key 2nd operand
4.	+	X: 5	Drop stack & compute

EXPECTED LCD X: 5 counters

1 **Place 2 counters on frame. Add 1 more counter.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Place 3 counters on frame. Add 2 more counters.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **You have 4 counters. Remove 1 counter.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **What number represents an empty ten-frame?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 1

Numbers to 5: Counting & Cardinality • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Ten-Frame Studio (2x5 Grid) + Counters

CAD Blueprint: ten_frame_tray

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/lib/ten_frame_tray.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal & Off-White (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Place 2 counters on frame. Add 1 more counte** Standard: CCSS.MATH.K.CC.B.4	3	[2] [ENTER] [1] [+]
2	**Place 3 counters on frame. Add 2 more counte** Standard: CCSS.MATH.K.CC.B.4	5	[3] [ENTER] [2] [+]
3	**You have 4 counters. Remove 1 counter.** Standard: CCSS.MATH.K.CC.B.4	3	[4] [ENTER] [1] [-]
4	**What number represents an empty ten-frame?** Standard: CCSS.MATH.K.CC.B.4	0	[0]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Watch for students recounting from 1 instead of continuing from the current register value.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.